ORTHOPEDIC EXAM OF THE BACK AND LOWER EXTREMITY

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Objectives

- Organize a clinical approach to examine a patient with a musculoskeletal lower extremity complaint
- Describe the patient positioning, physician hand placement and motion being induced for each orthopedic test
- Describe the anatomy being stressed and what injury/pathology may be present with a positive test
- Demonstrate a thorough orthopedic exam of the lower extremity.

- Identify causes of low back pain (LBP)
- Perform an evaluation of a patient with LBP, including detailed trauma history
- Understand role of Duration (Acute/subacute/Chronic) to etiology
- Develop a differential diagnoses for LBP
- Classify clinical symptoms/signs of red flags for LBP and know when to refer the patient to a specialist
- Formulate an appropriate clinical treatment plan based upon history and physical exam including when to use Osteopathic Manipulative Treatment (OMT) and when to refer the patient to a specialist

THE BACK

How common is LBP?

- Primary Care physicians will see at least one case of low back per week
- 2011 data – 85% of people will experience Low back pain in their life
- 2011 data – 5th most common reason for all physician visits in US
- 2nd most common symptomatic reason next to headache (which is frequently caused by neck pain
- At any given time, 1% of the US population is disabled and another 1% is due to LBP

Back Pain

- Majority of people will improve without a physician interaction
- 30-60% recover within 1 week
- 60-90% recover within 6 weeks
- 95% recover in 12 weeks
- ***Remember this – as you consider your diagnostic decisions***

Unchangeable LBP Risk Factors

Individual	Anatomic
Increasing Age High Birth Weight Spinal Birth Defects	Degenerative Disc Disease
Male Gender	Osteoarthritis
Family History	Synovial Cyst Formation
Previous Back Injury	Transitional lumbar vertebrae
Pregnancy	Schmorl's nodes
Previous Spine Surgery	Annular Disruption
High Birth Weight	Spondylolysis Spondylolithesis

LBP Prevention

- Proper Lifting techniques
- Weight management
- Avoid use of back braces
- Frequent activity/back strengthening exercises/core strength

Condition	LBP Red Flags	Action
Cancer	History of Cancer Unexpected weight loss Age >50 Failure to improve with therapy Pain 4-6 weeks/night and/or rest pain	If malignant spine disease suspected, imaging, CBC and ESR Identify possible primary malignancy should be investigated (eg. PSA, mammogram, serum or urine protein electrophoresis)
Infection	Fever History of intravenous drug use Recent bacterial infection (eg. UTI, pneumonia, skin cellulitis) Immunocompromised	If infection of spine is suspected, MRI, CBC, ESR and/or Urinalysis are indicated
Cauda Equina Syndrome	Urinary incontinence or retention Saddle anesthesia Anosphincter tone decrease/fecal incontinence Bilateral Lower extremity weakness/numbness or regressive neurological deficits	Immediate surgical consultation
Fracture	Use of corticosteroids Age 70 or history of osteoporosis; recent significant trauma	Appropriate imaging and surgical consultation
Acute Abdominal Aneurysm	Abdominal pulsating mass Other atherosclerotic vascular disease, age 60, rest/night pain	Appropriate imaging (ultrasound) and surgical consultation
Significant herniated nucleus pulposis (HNP)	Major muscle weakness	Appropriate imaging and surgical consultation

Maybe remove ESR for Cacner

Add alcohol and history of fall

Yellow Flags

Signs for those who may fail to improve timely or worsen with management/treatment

Attitudes & Beliefs	Comorbid Conditions
Fear of movement; use of excessive rest Catastrophizing Excessive focus on pain Feeling out of control Passivity toward rehabilitation	Lack of or decreased sleep History of other disabling injuries or conditions Lack of financial incentive to return to work (secondary gain) Pending litigation Overprotective family
Affective factors Poor work history Poor compliance with rehabilitation Withdrawal from activities (ADLs, social) History of substance abuse (self-medicating) Depression Anxiety Irritability History of psychological or physical abuse History of irregular or episodic physical activity Conflicting diagnosis from different providers	Waddell signs Pain with axial loading of spine Superficial tenderness to palpation (light touch) Overreaction (pain out of proportion to physical findings) Straight leg raise test improves with distraction Regional weakness

Waddell very important

Modifiable LBP Risk Factors

Individual	Psychosocial	Occupational	Other
Sedentary Life Style	Stress	Unemployment	Poor Pain Tolerance
Smoker	Depression	Perceived Inadequate Income	History of Recurrent Back Pain
Overweight/Obese	Decreased Cognition	Monotonous Tasks	Presence of Sciatica
Poor Posture & Biomechanics	Somatization	Long Periods of Sitting/Standing	Corticosteroid Use
Education about managing back pain	Fear/Avoidance Behaviors	Heavy Lifting	Chronic Illness Causing Chronic Cough (eg. COPD)
Poor General Health	History of Anxiety	Repetitive bending, twisting, squatting, pushing, pulling	Participation strenuous and/or contact sports

History

- Low Back – **get thorough history**
  - **Mechanism of injury – details, details, details**
    - ”What were you doing when the pain started?”
    - ”How did it happen?”
    - Patients may say, “Don’t know…just woke up with it”
  - **Exact location of the pain**
    - Specific vertebrae, regional pain, soft tissues of back

- **Duration & timing –** acute, subacute, insidious
- **Character –** ache, dull, stabbing, burning, numbness
- **Comorbid conditions**
- **Radicular symptoms (Radiation)**
- **Aggravating and Relieving Factors**
- **Hx of prior injury/Pain –** include prior work up info and treatments, time frames for the pain

AXIAL OR RADICULAR

Back Physical Exam

- Inspection
  - Exposure to the Lumbosacral spine, hips, thighs, abdomen and pelvis(if indicated)
  - Note curvatures
- Palpation
  - Include the spinous processes of the L-S spine, hips, pelvis, sacroiliac joints, paraspinous muscles, piriformis muscles, hamstrings and quadriceps
  - Includes your osteopathic structural examination
  - Must ask patient if this is tender/painful

- Range of Motion (ROM) – active and passive
  - Is ROM associated or limited by pain?
  - Is pain changed by flexion or extension?
- Radicular (neurologic) testing, including straight leg raising (to r/o nerve root impingement)
- Gait

Neurologic Testing for LBP

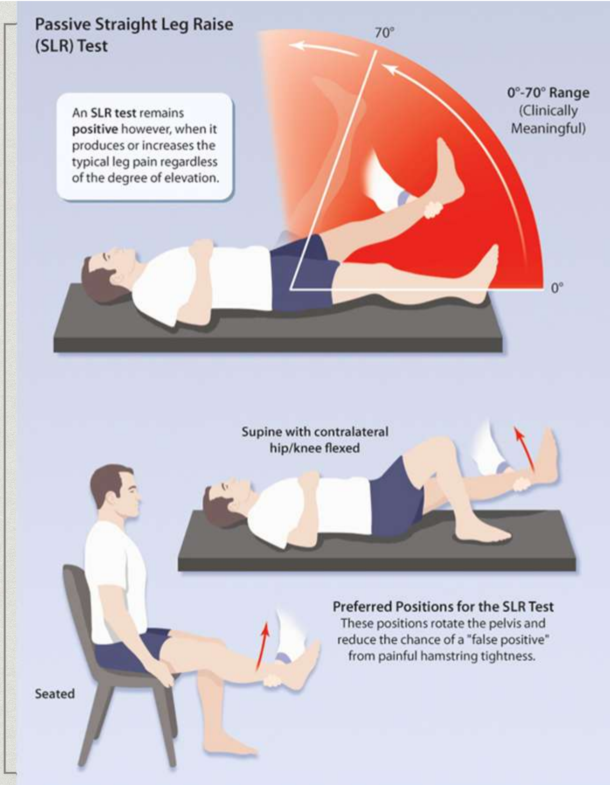
- Deep Tendon Reflexes
  - Achilles
  - Patellar
- Muscle (strength) testing
  - Ankle and great toe dorsiflexion (L4-L5)
  - Plantar flexion (S1)
- Sensory testing (Light touch)
  - Medial foot (L4)
  - Dorsal Foot (L5)
  - Lateral foot (S1)

This is review!

Straight Leg Raising (SLR)

- **Supine**
  - To Perform: Patient is supine; passive flexion of the extended leg at the hip (stretches L5 and s1 nerve root & the sciatic nerve); passive dorsiflexion of the foot further stretches the nerve root
  - Positive: Maneuver reproduces the lower back and/or limb pain
  - Normal for hamstring tightness and or lower back discomfort at 80^0
- **Crossed Leg**
  - To Perform: Same as supine
  - Positive: when Usual pain is reproduced in the opposite side Limb or buttock
- **Seated**
  - To Perform: Patient is seated with knees bent at 90^0; extend lower leg at the knee and dorsiflex the foot.
  - Positive: reproduces the pain distal to the knee

High sensitivity and low specificity : in my experience only for low lumbar pathology



The key to SLR: the maneuver reproduces the patient's usual pain

[SLR Video]

FEMORAL NERVE STRETCH

Femoral Nerve Stretch Test

Femoral Nerve Stretch Test (or Ely test) is a way of eliciting root stretch in the evaluation of high lumbar radiculopathy.

** UPPER radiculopathies **

LBP Differential Diagnosis

- Myofascial pain syndrome
- Lumbosacral strain and/or sprain
- Degenerative disc disease
- Lumbar spondylosis
- Lumbar spinal stenosis
- Lumbar radiculopathy (common term is “sciatica”)
- Lumbar spine fracture
- Sacroiliitis
- Somatic dysfunction

Changed

Don’t forget to r/o hip and pelvis etiologies, as well as urinary tract and pelvic pain symptoms

Always remember to consider malignancies – primary or metastatic (night pain, fever, weight loss should always be red flags)

Imaging

- Plain film x-rays – indicated if there is trauma; occupational guidelines – no plain films for 6 weeks
- MRI – neurologic signs identified with no response to conservative treatment OR if there is a red flag (cauda equina, cancer, cervical fracture) – done with gadolinium contrast
    - **NOT first line order if you don't have a red flag**
- CT scans – usually done with iodine contrast and myelogram is added – no response to conservative treatment
- Ultrasound – to r/o AAA or other intra-abdominal process presenting as back pain

Advanced imaging if no improvement for 6 months*

LBP Lab Testing

- CBC, ESR
- ANA, Rheumatoid factor
- Urinalysis

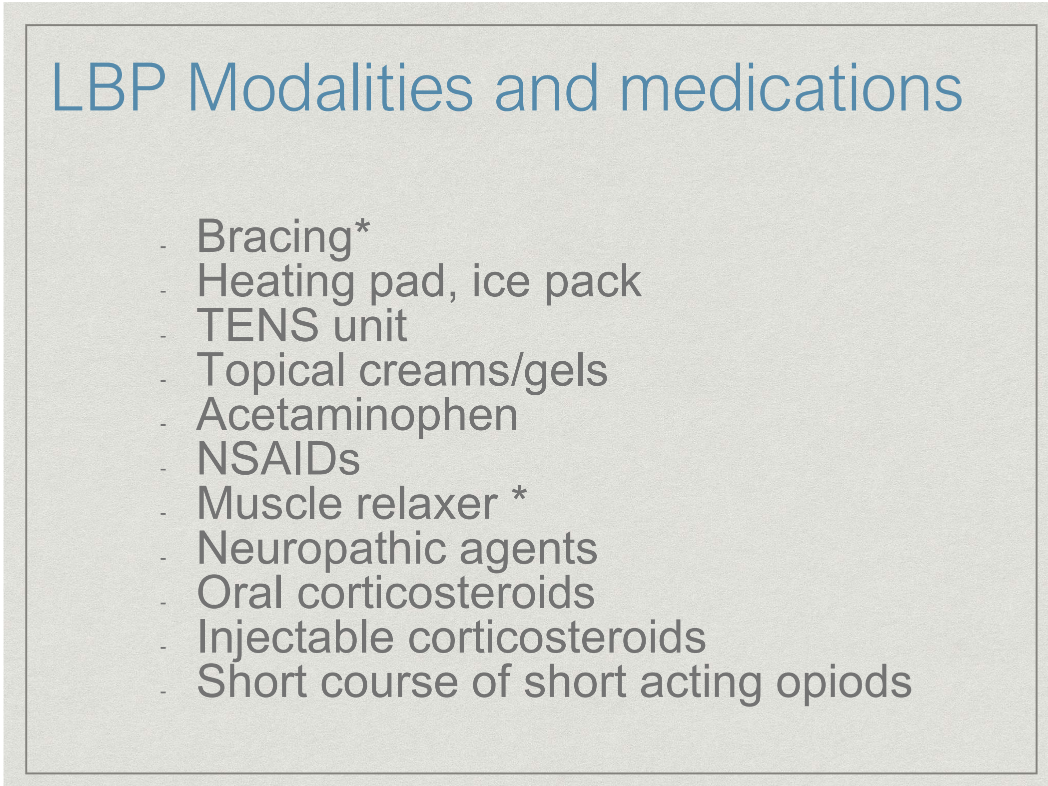
LBP Other Diagnostic Tools

- EMG/NCV (electromyography and nerve conduction velocities) – more commonly used to diagnose carpal tunnel or tarsal tunnel syndromes
- DXA (Bone density test) - osteoporosis
- Discogram – very painful; little evidence to support this

4 pillars of low back treatment

- 1. Ergonomic optimization
- 2. Home exercises or physical therapy
- 3. Modalities and Medications
- 4. Interventional procedures

Addition



Start with heat, ice, gentle stretching,

Osteopathic Treatment

- HVLA
- Counterstrain
- Facilitated Positional Release
- Soft Tissue
- Myofascial Release

- ***Any are options – you should be able to give specific treatment plans based on the findings***

Patient Education

- Exercises for home
- Proper sleep – cervical pillow, pillow between knees
- Yoga
- Limitations for job if indicated
- Limitations for bedrest
- Realistic expectations for recovery
- Follow up – depends on acuity

Consider removing

Take Away

Axial or Radicular?

Red Flags?

Optimize minimal invasive treatment first

Fictional improvement > Pain control

LOWER EXTREMITY

Overview

- Presentation of a lower extremity complaint
- How to take a thorough history of a lower extremity complaint
- Anatomy review, physical exam and special tests of:
    - Hip
    - Knee
    - Ankle and Foot

Presentation

- Pain
- Instability
- Decreased range of motion
- Weakness
- Change in gait
- Inability to perform ADL's

History

- OLD CARTS
    - Onset
    - Location/Radiation
    - Duration
    - Character
    - Aggravating factors
    - Relieving factors
    - Timing
    - Severity
- What is the functional limitation?
- Do symptoms affect a single region or multiple joints?
- Acute onset or slowly progressive?
- If injury, what was the mechanism of injury?
- Prior problems with the affected area?
- Systemic symptoms?
- Pain with certain movements?
- Therapies tried?

More History

- PMH
  - Osteoarthritis, Rheumatoid arthritis, Autoimmune disease, Gout, Systemic illness, Diabetes, Heart disease
- PSH
  - Orthopedic surgeries, other recent surgeries
- FH
  - Arthritis, Gout, Autoimmune diseases, Diabetes, Heart disease
- SH
  - Work, Hobbies, Exercise, History of STIs, Current or past IV drug use

Clinical Pearls for Orthopedic Testing of the LE

- Know your anatomy!
- Inspect, inspect, inspect
  - What does their gait look like?
  - How do they move?
  - Are they able to get on the exam table unassisted?
  - Are they using an assistive device (cane, walker, brace)?
  - Is there muscle atrophy?
- Active ROM always precedes passive ROM
- Examine both sides
- Test the unaffected side first

To Sprain or to Strain, that is the question

- **Sprain**: injury to ligament or joint capsule
  - Ligaments attach bone to bone
  - Acute injury that result from trauma that displaces the surrounding joint from its normal alignment
  - Bruising, swelling, instability and painful movement are common
- **Strain**: injury to a tendon or muscle
  - Tendons attach muscles to bones
  - Acute injury that results from over stretching or over contraction
  - Pain, weakness, and muscle spasms are common

Grading Sprains

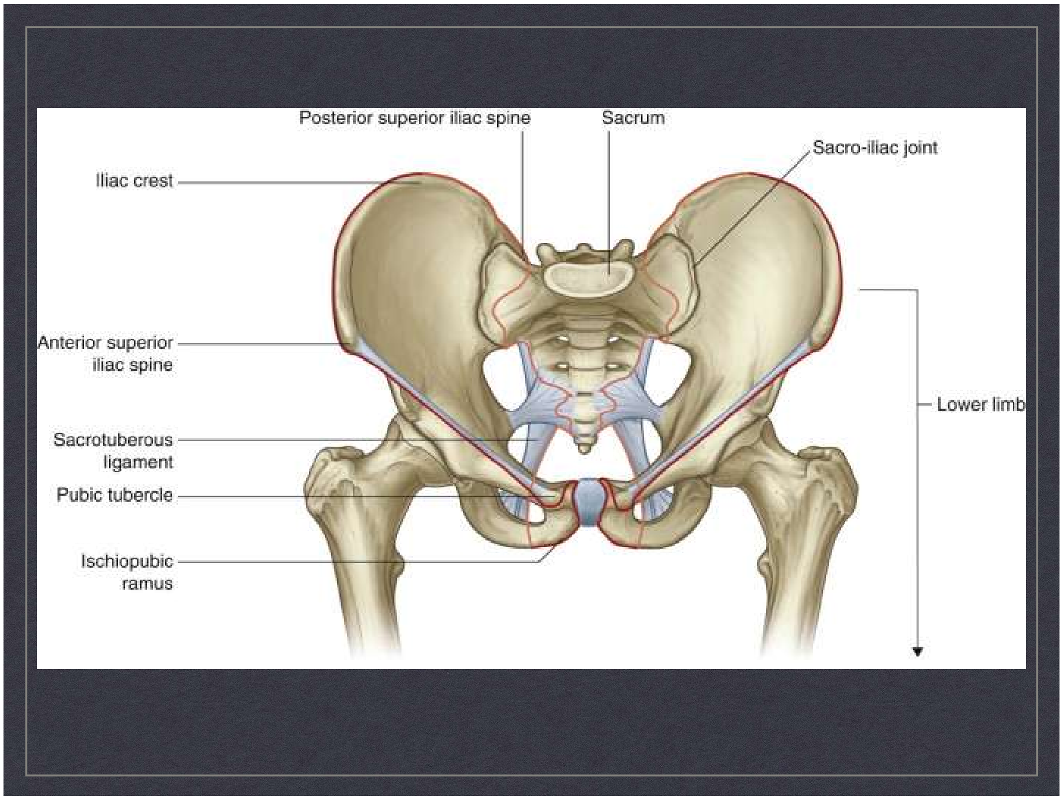
- Grade 1: A few torn ligamentous fibers
  - Mild joint pain reproduced on local palpation
  - Firm endpoint with **minimal laxity**
  - No joint instability
- Grade 2: Increased joint instability
  - Joint swelling and pain
  - Firm endpoint with **some joint laxity**
  - Decreased function
- Grade 3: Complete ligamentous tear
  - Minimal or no endpoint, **gross joint laxity**
  - Unstable joint
- **Strains** are graded with the same terminology for muscles

THE HIP

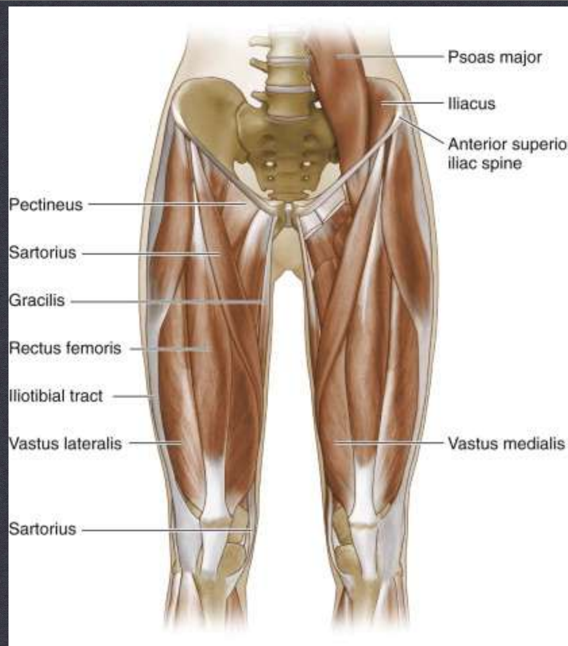
ORTHO EXAM

DATE 2016

CLIENT

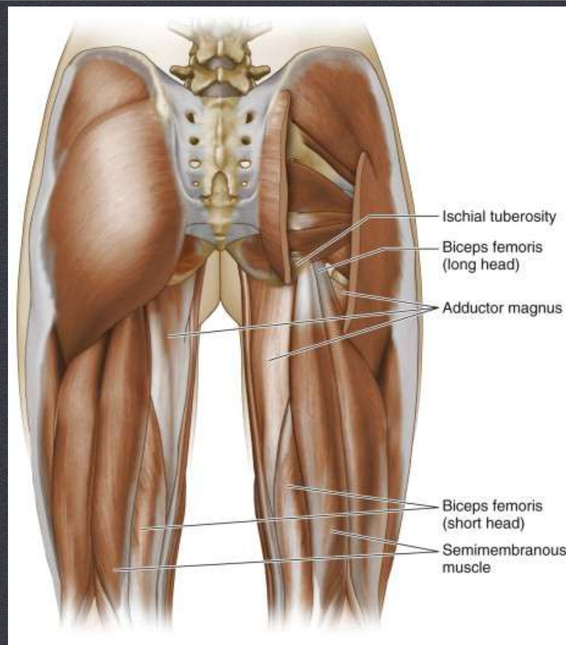


Posterior superior iliac spine
Sacrum
Sacro-iliac joint
Iliac crest
Anterior superior iliac spine
Lower limb
Sacrotuberous ligament
Pubic tubercle
Ischiopubic ramus



ANATOMY OF THE HIP

ANTERIOR



ANATOMY OF THE HIP

POSTERIOR

Physical Exam

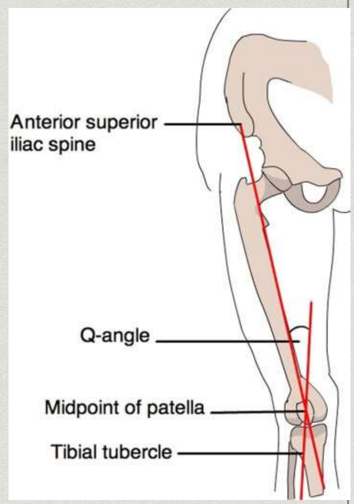
- Inspection
- Gait
- Palpation
  - ASIS
  - Greater Trochanter
- Range of Motion
  - Active
  - Passive
- Muscle strength
- Provocative tests

Inspection of Hips

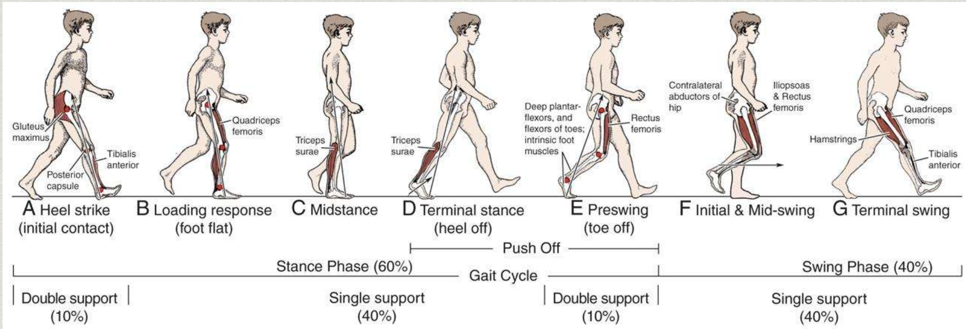
- Look for overall alignment of hip, knee, and ankle
- Look for muscular atrophy
- Note if iliac crests are equal or one is lower/higher

Q angle

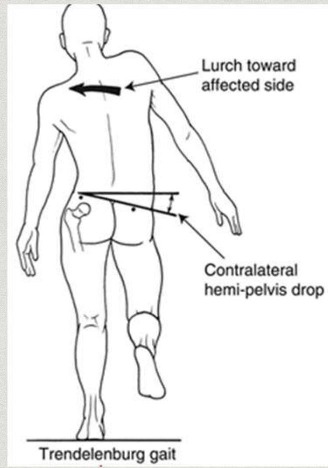
- Formed by a straight line from ASIS to patella, then a second line from the midpoint of the patella to the tibial tubercle. The angle formed from the intersection of these lines is the Q angle.
- Men have an average Q angle 15 or less
- Women have an average Q angle of 20 degrees or less (wider pelvis width)
- This plays a role in knee ligamentous injuries and quadriceps muscle strength to support the knee.



Gait Analysis



Trendelenburg Gait

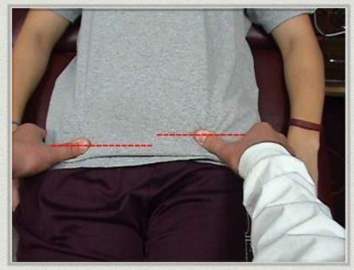


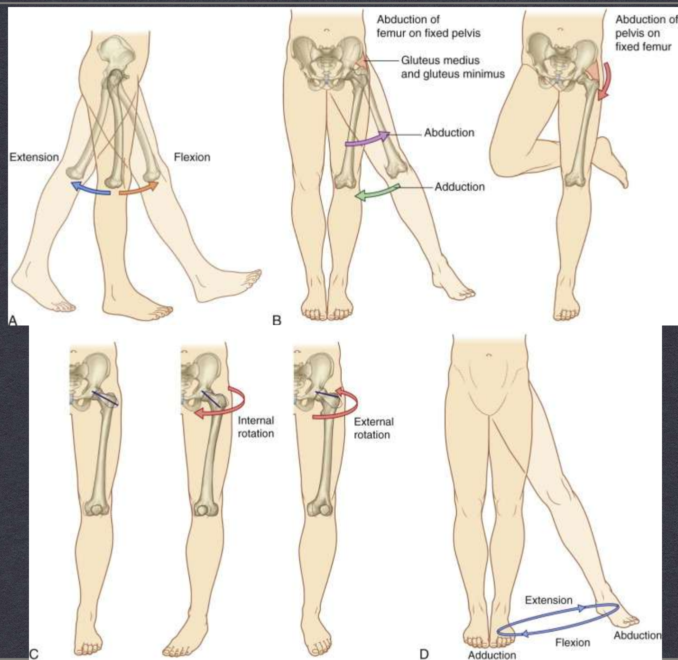
[Trendelenburg Video]

ABNORMAL GAIT CAUSED BY WEAKNESS OF THE ABDUCTOR MUSCLES OF THE LOWER LIMB, GLUTEUS MEDIUS AND GLUTEUS MINIMUS.

Palpate

- Palpate ASIS
  - Evaluate for symmetry and tenderness
    - Tenderness may indicate avulsion of sartorius or rectus femoris or meralgia paresthetica
- Palpate greater trochanter
  - Tenderness may indicate greater trochanteric bursitis or gluteus medius tendinitis





RANGE OF MOTION OF THE HIP

Active Range of Motion

To test active range of motion, ask the patient to do a squat to a comfortable level.

Passive Range of Motion

Test with the patient supine

Hip Flexion

Flex hip until pelvis begins to rotate. Normal hip flexion in adults is 110º to 130º

Internal Rotation

External Rotation

Internal and External Rotation

Flex hip to 90º and with the thigh perpendicular to the ASIS rotate the tibia away from midline to measure internal rotation of the hip and toward midline to measure external rotation of the hip.

Muscle Testing

Hip Flexors

Hip Extensors

Hip Abductors

Hip Adductors

Special Tests: Thomas Test

- Evaluates for hip flexion contracture or psoas tightness
- Have the patient lie supine with legs extended so the table edge does not contact the posterior calves
- Ask the patient to pull one hip into maximum flexion
- Observe the contralateral hip to see if it also flexes off the table
- A **NORMAL** hip will remain on the table or flex very slightly

Thomas Test Video

Abnormal Thomas Test

Special Tests: FABER Test

- Stands for **F**lexion-**AB**duction-**E**xternal-**R**otation
- This is a stress maneuver to detect hip and sacroiliac pathology
- With the patient supine, place the affected hip in flexion, abduction, and external rotation with the patient's foot on the opposite knee
- Stabilize the pelvis with your hand on the contralateral ASIS and press down on the affected side.
- A **POSITIVE** test is pain with the maneuver indicating the hip or sacroiliac joint may be affected.

FABER Test Video

Special Tests: Ober's Test

- Tests for iliotibial band tightness
- Pt is side lying with test side up
- Knee can be straight or at 90 degrees
- Hip is maintained in slight extension
- Leg is abducted then lowered toward the table with the pelvis stabilized
- Abnormal if unable to adduct to parallel with examining surface

Ober's Test Video

Hip Exam Video

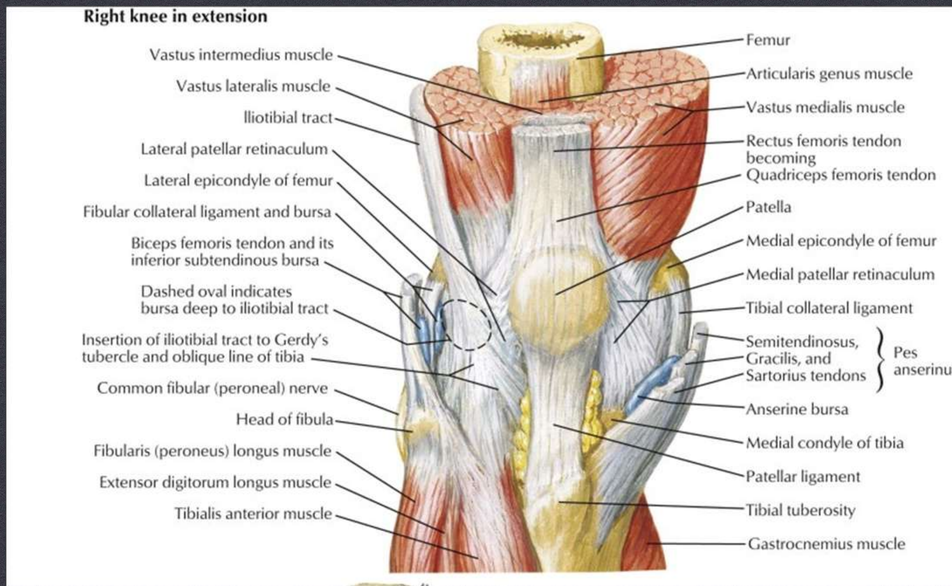
- Bates Hip Exam Video

THE KNEE

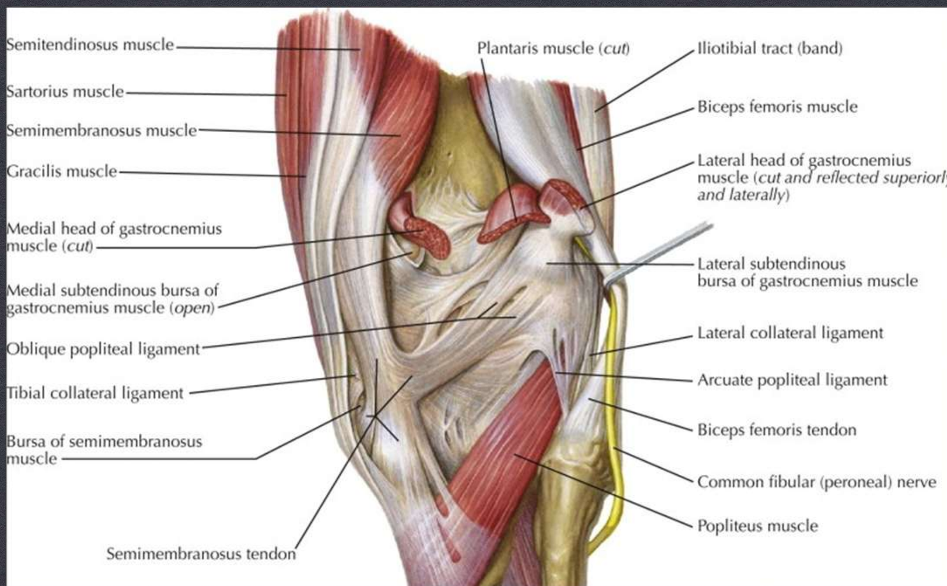
ORTHO EXAM

DATE 2016

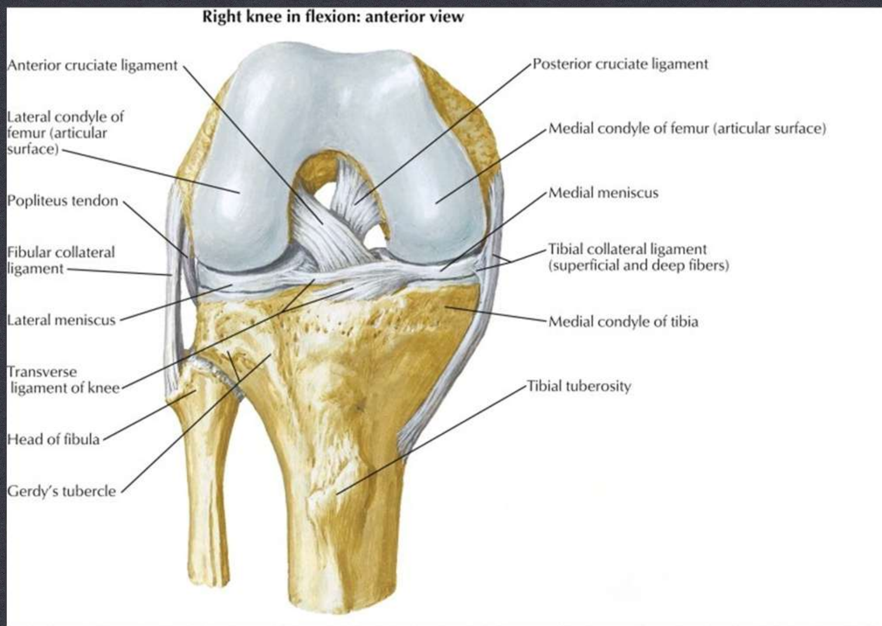
CLIENT



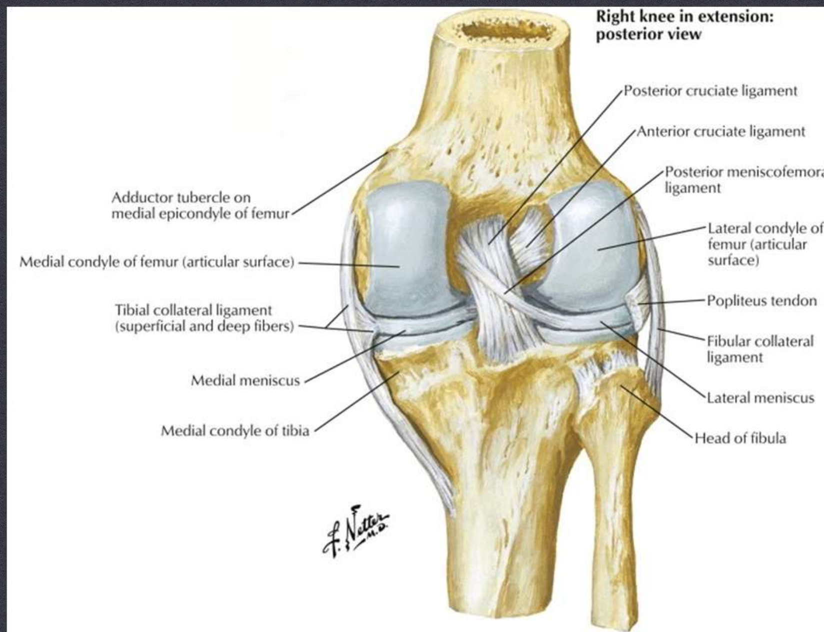
Right Anterior Knee in Extension



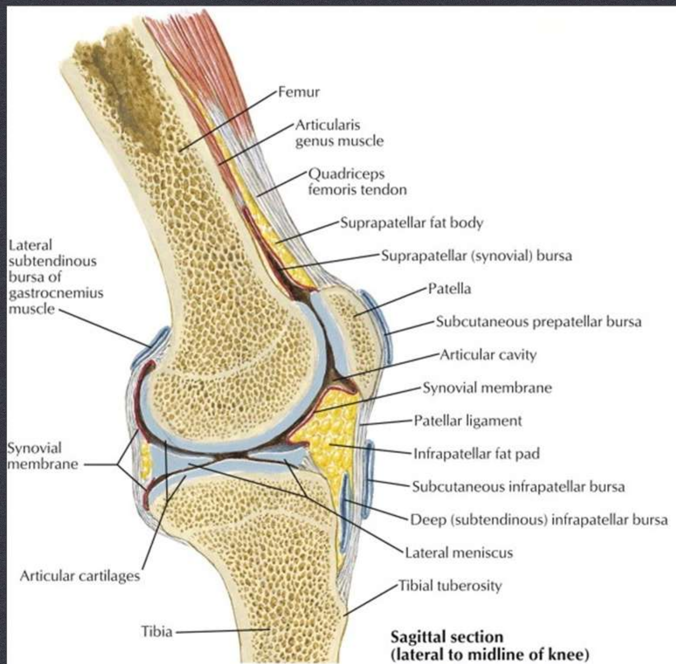
Right Posterior Knee in Extension



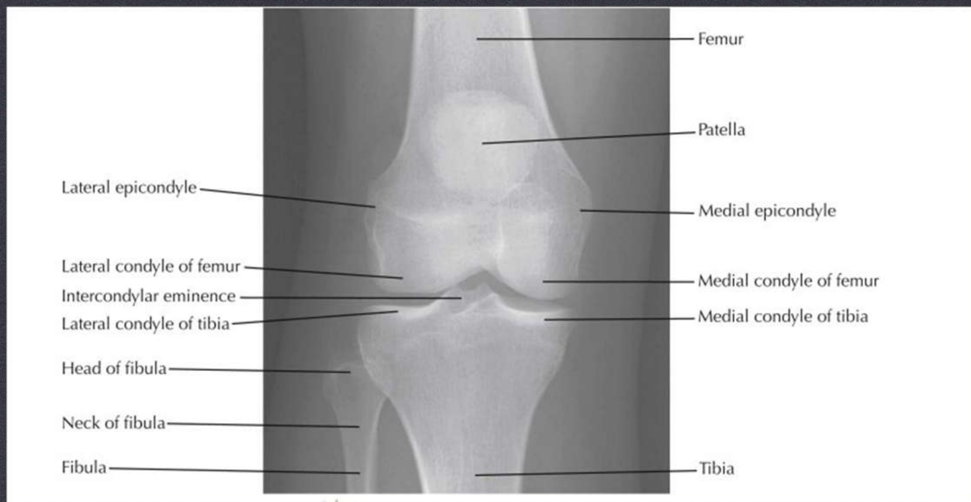
Right Anterior Knee in Flexion



Right Posterior Knee in Extension



Sagittal Section Lateral to Midline of Knee



Normal Xray of Right Knee

Physical Exam

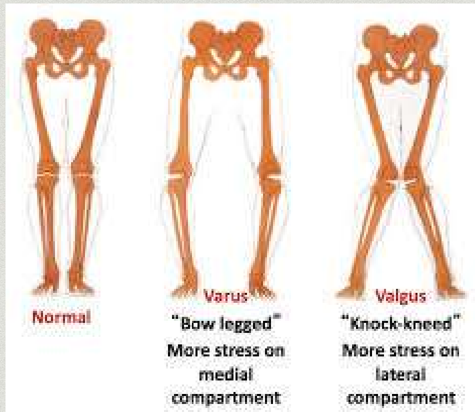
- Inspection
- Palpation
  - Knee Effusion
  - Patella
  - Joint Line Tenderness
- Range of Motion
  - Active- Already Assessed with Squat
  - Passive
- Muscle strength

- Provocative Tests
  - Patellar Apprehension Sign
  - Patellar Grind Test
  - Observe for Tibial Sag Sign
  - Anterior Drawer
  - Posterior Drawer
  - Lachman's Test
  - Valgus Stress at 0º and 30º
  - Varus Stress at 0º and 30º
  - McMurray's Test in external rotation
  - McMurray's Test in internal rotation

Inspection

- Anterior View:
  - Inspect with the patient standing.
  - Look for valgus (knock-knee) or varus (bowleg) deformities, asymmetry of alignment, and quadriceps atrophy.
  - Look for internal femoral torsion which will present as inward facing patellae when the feet are facing forward
- Posterior View:
  - Look for atrophy of the hamstrings or calf muscles.

Review of Varus and Valgus



Palpation: Knee Effusion

- Inspect the suprapatellar region for large effusions
- “Milk down” joint fluid
  - Physician’s superior hand starts 2 inches above the suprapatellar pouch and pushes inferiorly in a milking motion.   Hold the fluid wave in place
  - With the other hand, push straight down on the patella.
  - Excessive fluid will create a spongy feeling as the patella is pushed down.

Palpation: Patella

- Palpate the superior and inferior poles of the patella
  - May be tender superiorly in quadriceps tendinitis
  - May feel a palpable defect superiorly with quadriceps tendon rupture or inferiorly with patellar tendon rupture
- Displace the patella laterally and medially
  - Palpate the edge and undersurface

Palpation: Joint Line

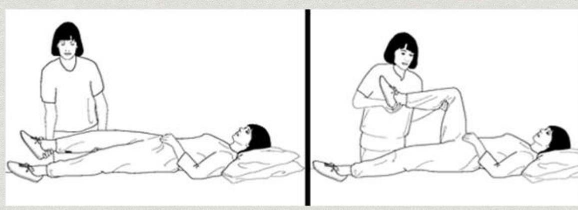
- Flex the knee to 90º and identify the joint line which is the soft spot between the femur and tibia
- Palpate along the joint margin on both the medial and lateral sides of the knee
- Tenderness supports a diagnosis of a torn meniscus

Range of Motion: Active

- Already assessed by asking the patient to do a squat

Range of Motion: Passive

- May be done supine or prone
- Examiner’s hands are place on the ankle and knee to flex and extend knee.
- Extension may also be measured by elevating the heel 5cm above the table while the patient is supine



Muscle Testing

- Ask a seated patient to extend knee and resist force pushing down on tibia to assess quadriceps
- Ask a seated patient to flex knee and resist force pulling out on tibia to assess hamstrings

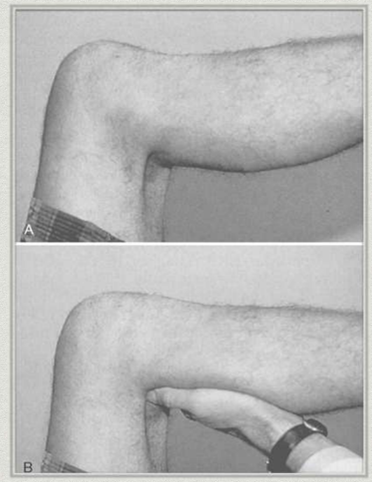
Patellar Apprehension Sign

- With the patient supine and the knee relaxed to about 30º of flexion, use your thumbs to displace the patella laterally
- Normal test: Examiner will be able to move the patella laterally before reaching an end point
- Abnormal test: Absence of firm end point. Patient may anticipate or perceive pain and becomes apprehensive or contracts the quadriceps to avoid further lateral displacement. Indicates **patellar instability.**

Patellar Grind Test

- With the patient supine and knee extended, place one hand superior to the patella and push the patella inferiorly.
- Ask the patient to tighten the quadriceps against this patellar resistance.
- A grinding sound and/or pain may indicate **patellofemoral chondromalacia**
- This test is often positive in normal patients

Tibial Sag Sign



- Examiner holds the knee and hip in 90º of flexion.
- Observe the proximal tibia just inferior to the anterior joint line.
- The tibia will “drop back” or sag on the femur if the **Posterior Cruciate Ligament is torn**

Anterior Drawer Test

- Patient is supine and knee flexed to 90º, hip flexed to 45º and in neutral rotation
- Examiner stabilizes the leg by sitting on the patient’s foot
- Palpate the hamstring tendons to be sure they are relaxed
- Grasp the proximal tibia with both hands and slide the tibia anteriorly
- Positive test is laxity of the ligament compared to the unaffected side
- Assesses the **Anterior Cruciate Ligament**

Posterior Drawer Test

- Patient is supine and knee flexed to 90º, hip flexed to 45º and in neutral rotation
- Examiner stabilizes the leg by sitting on the patient's foot
- Palpate the hamstring tendons to be sure they are relaxed
- Grasp the proximal tibia with both hands and push the tibia posteriorly
- Positive test is laxity of the ligament compared to the unaffected side
- Assesses the **Posterior Cruciate Ligament**

Anterior and Posterior Drawer Tests

- Can be repeated with internal rotation and external rotation to further isolate the structures of the knee
    - Internal rotation: induces PCL and posteriorlateral joint capsule
    - External rotation: induces ACL and posteriomedial joint capsule and MCL

Lachman Test

- Used to assess the **Anterior Cruciate Ligament**
- Patient is supine and knee flexed to 30º.
- Support the thigh so muscles are relaxed. This may be done with the examiner's knee under the thigh.
- Examiner grasps the distal femur from the lateral side and the proximal tibia from the medial side with the other hand
- Lift or pull the tibia anteriorly while stabilizing the femur
- Abnormal test: increased anterior tibial translation and/or absence of a firm end point indicate a partial or complete tear of the ACL

Valgus Stress Test

- Assesses the **Medial Collateral Ligament**
- Patient is supine and thigh is supported to relax quadriceps.
- Place one hand on the lateral side of the knee. Grasp the medial distal tibia with the other hand and ABduct the knee.
- Perform in extension and at 30º of flexion
- Abnormal test:
    - At 30º of flexion, the affected knee opens up in a valgus direction more than the unaffected side indicating an injury to the MCL
    - In extension, if the affected knee opens more than the unaffected side, the patient has a severe injury involving the MCL, PCL and possibly other structures

Varus Stress Test

- Assesses the **Lateral Collateral Ligament**
- Patient is supine. Place one hand on the medial side of the knee and grasp the lateral distal tibia with the other hand. ADduct the knee.
- Perform in extension and at 30º flexion
- Abnormal test:
  - At 30º flexion, if the affected knee opens up more than the unaffected knee, there is a partial or complete tear of the Lateral Collateral Ligament
  - In extension, if the affected knee opens up more than the unaffected knee, the patient has a severe injury involving more than the LCL.

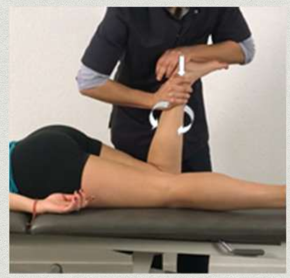
McMurray Test

- Evaluates the knee menisci
- Patient is supine. Flex the knee to the maximum pain-free position
- Hold the leg in that position while eternally rotating the foot (A)
- Gradually extend the knee while maintaining the tibia in external rotation (B)
  - This will stress the **Medial Meniscus**
- Same maneuver is performed while rotating the foot internally
  - This will stress the **Lateral Meniscus**
- Abnormal test: Causes click or pain in patients with a tear in the posterior horn of indicated meniscus

Apley Compression and Distraction Test

-Appley's grinding test involves placing the patient in the prone position with the knee flexed to 90 degrees .

-The patient's thigh is then rooted to the examining table with the examiner's knee. The examiner laterally and medially rotates the tibia, combined first with distraction, while noting any excessive movement, restriction or discomfort.



-The process is then repeated using compression instead of distraction

-If rotation plus distraction is more painful or shows increased rotation relative to the normal side, the lesion is most likely to be **ligamentous** If the rotation plus compression is more painful or shows decreased rotation relative to the normal side, the lesion is most likely to be a **meniscus** injury.

Whoa…That was a lot. Let's review what tests what

- Anterior Cruciate Ligament (ACL)
  - Anterior Drawer
  - Lachman Test
- Posterior Cruciate Ligament (PCL)
  - Posterior Drawer Test
  - Tibial Sag Sign
- Lateral Collateral Ligament (LCL)
  - Varus Stress Test
- Medial Collateral Ligament (MCL)
  - Valgus Stress Test
- Medial Meniscus
  - McMurray's in external rotation
  - Medial joint line tenderness
- Lateral Meniscus
  - McMurray's in internal rotation
  - Lateral joint line tenderness

Knee Exam Video

- Bates Knee Exam Video

Additional Knee Exam Videos

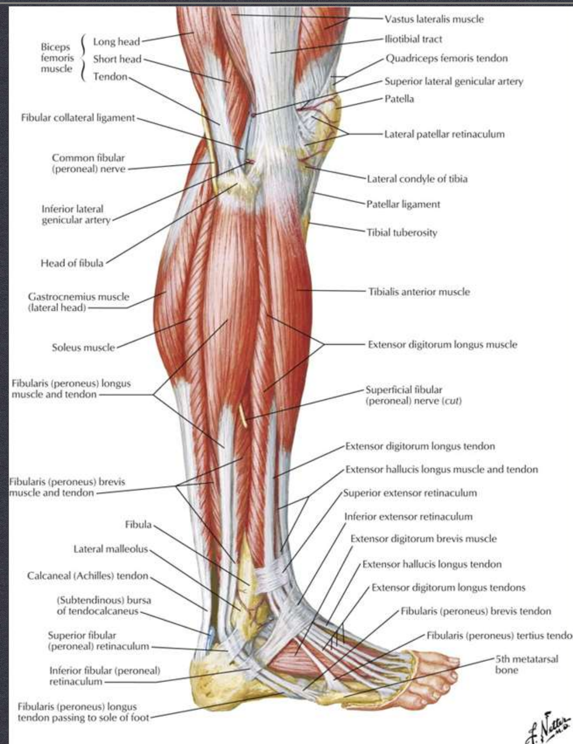
- Knee Exam
- This is a great series of videos of the knee exam by the British Journal of Sports Medicine

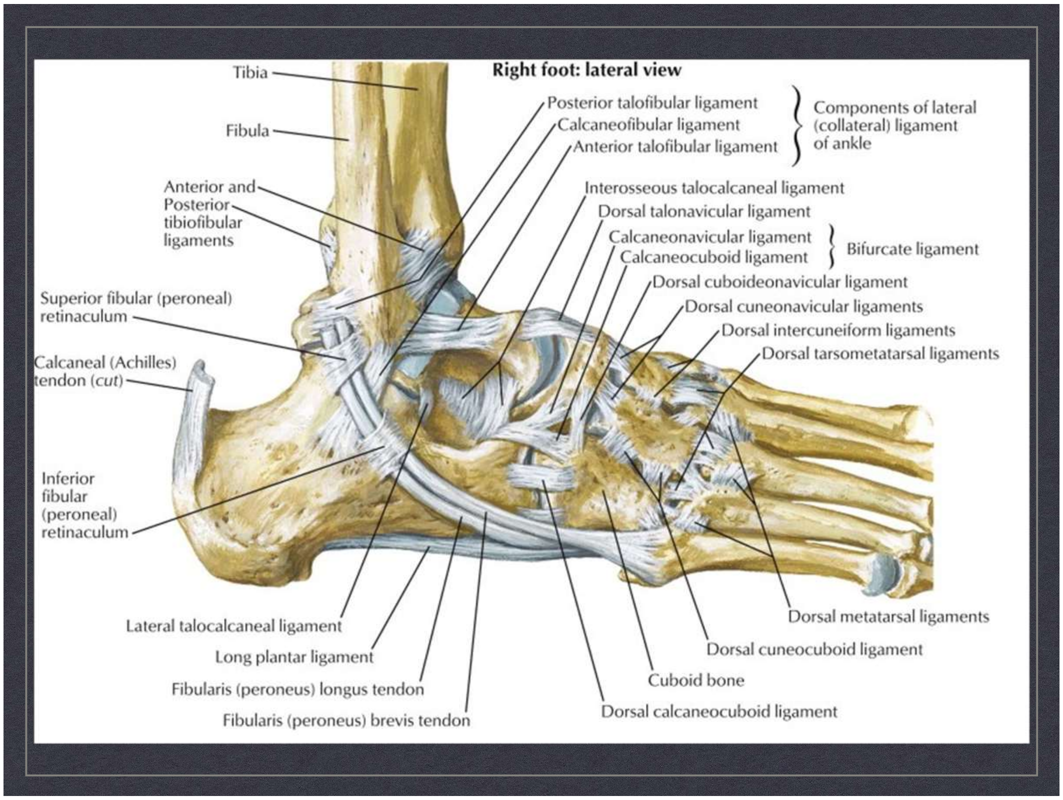
THE ANKLE

ORTHO EXAM

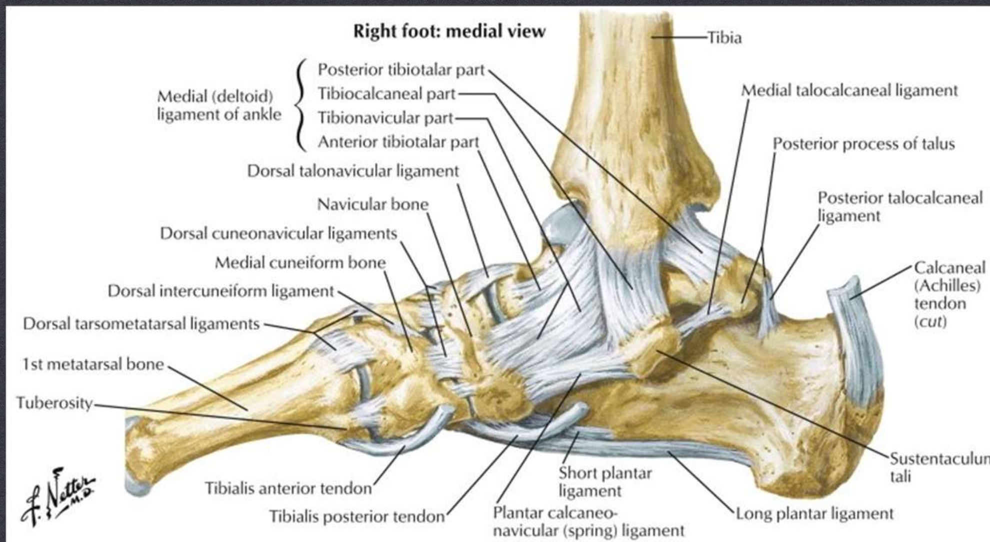
DATE 2016

CLIENT

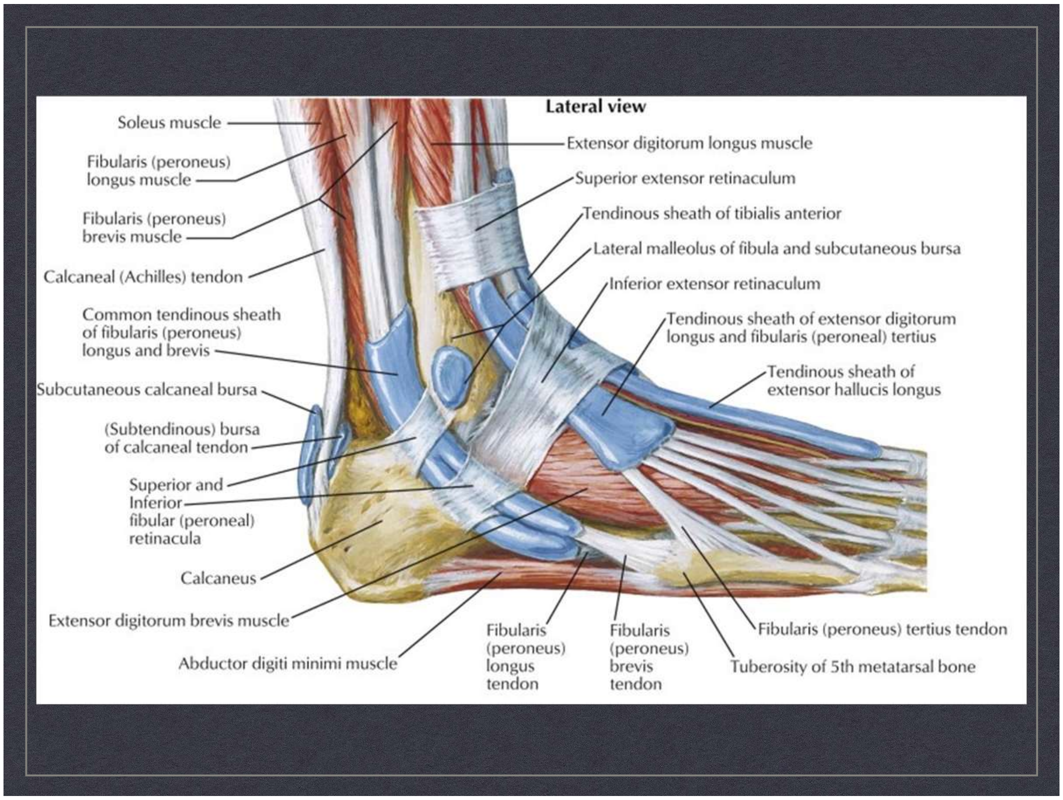

Vastus lateralis muscle
Biceps femoris muscle
Long head
Short head
Tendon
Iliotibial tract
Quadriceps femoris tendon
Superior lateral genicular artery
Patella
Fibular collateral ligament
Lateral patellar retinaculum
Common fibular (peroneal) nerve
Lateral condyle of tibia
Inferior lateral genicular artery
Patellar ligament
Tibial tuberosity
Head of fibula
Tibialis anterior muscle
Gastrocnemius muscle (lateral head)
Soleus muscle
Extensor digitorum longus muscle
Fibularis (peroneus) longus muscle and tendon
Superficial fibular (peroneal) nerve (cut)
Extensor digitorum longus tendon
Extensor hallucis longus muscle and tendon
Fibularis (peroneus) brevis muscle and tendon
Superior extensor retinaculum
Inferior extensor retinaculum
Fibula
Extensor digitorum brevis muscle
Lateral malleolus
Extensor hallucis longus tendon
Calcaneal (Achilles) tendon
Extensor digitorum longus tendons
(Subtendinous) bursa of tendocalcaneus
Fibularis (peroneus) brevis tendon
Superior fibular (peroneal) retinaculum
Fibularis (peroneus) tertius tendon
Inferior fibular (peroneal) retinaculum
5th metatarsal bone
Fibularis (peroneus) longus tendon passing to sole of foot



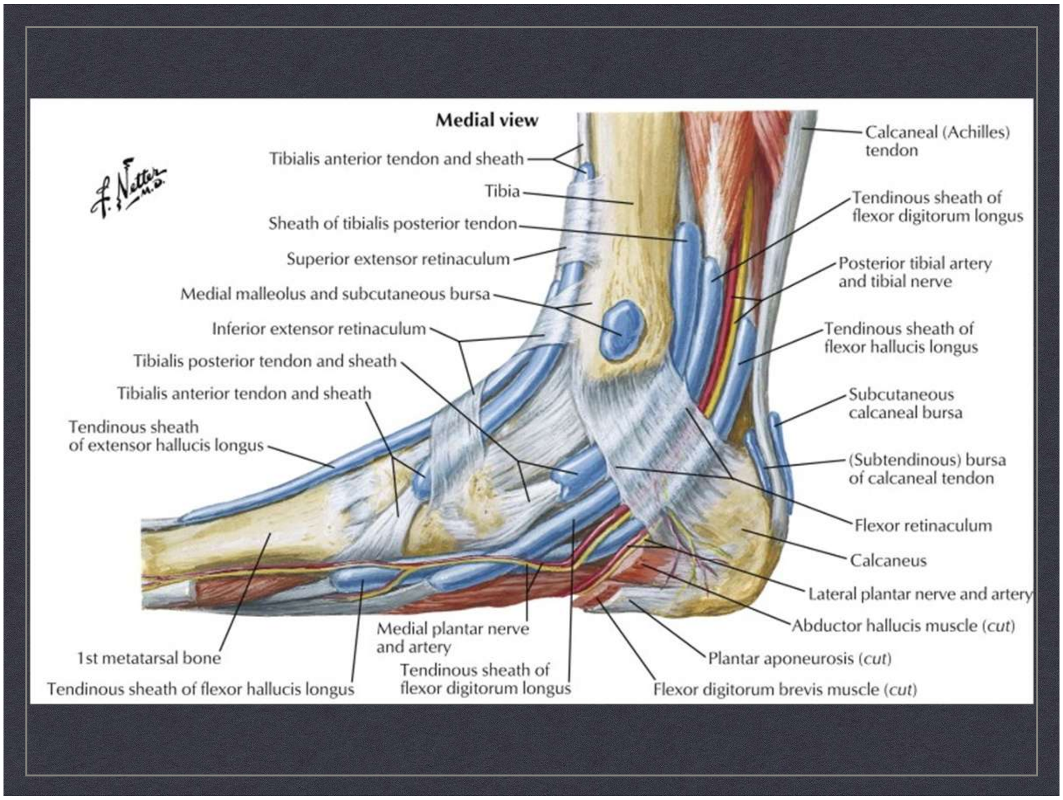
Right foot: lateral view
Tibia
Fibula
Anterior and Posterior tibiofibular ligaments
Superior fibular (peroneal) retinaculum
Calcaneal (Achilles) tendon (*cut*)
Inferior fibular (peroneal) retinaculum
Posterior talofibular ligament
Calcaneofibular ligament
Anterior talofibular ligament
Components of lateral (collateral) ligament of ankle
Interosseous talocalcaneal ligament
Dorsal talonavicular ligament
Calcaneonavicular ligament
Calcaneocuboid ligament
Bifurcate ligament
Dorsal cuboideonavicular ligament
Dorsal cuneonavicular ligaments
Dorsal intercuneiform ligaments
Dorsal tarsometatarsal ligaments
Dorsal metatarsal ligaments
Dorsal cuneocuboid ligament
Cuboid bone
Dorsal calcaneocuboid ligament
Lateral talocalcaneal ligament
Long plantar ligament
Fibularis (peroneus) longus tendon
Fibularis (peroneus) brevis tendon



Right foot: medial view
Tibia
Medial (deltoid) ligament of ankle
Posterior tibiotalar part
Tibiocalcaneal part
Tibionavicular part
Anterior tibiotalar part
Medial talocalcaneal ligament
Posterior process of talus
Dorsal talonavicular ligament
Navicular bone
Posterior talocalcaneal ligament
Dorsal cuneonavicular ligaments
Medial cuneiform bone
Calcaneal (Achilles) tendon (cut)
Dorsal intercuneiform ligament
Dorsal tarsometatarsal ligaments
1st metatarsal bone
Tuberosity
Sustentaculum tali
Short plantar ligament
Tibialis anterior tendon
Tibialis posterior tendon
Plantar calcaneo-navicular (spring) ligament
Long plantar ligament

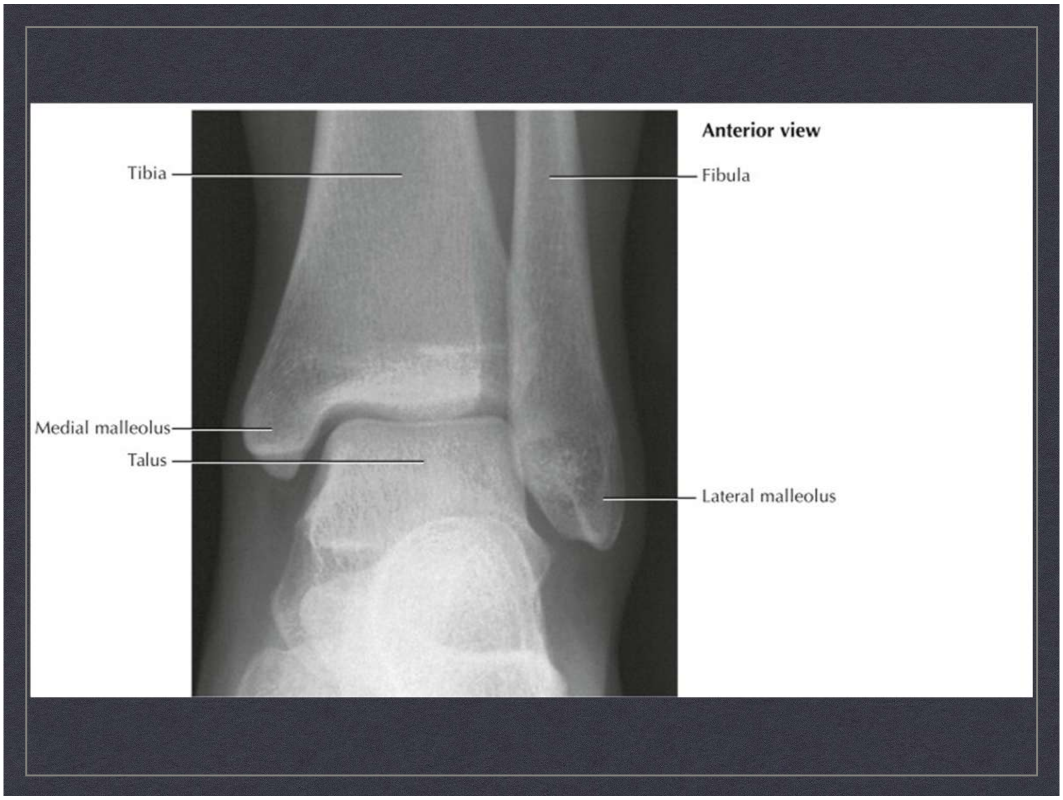


Lateral view
Soleus muscle
Fibularis (peroneus) longus muscle
Fibularis (peroneus) brevis muscle
Calcaneal (Achilles) tendon
Common tendinous sheath of fibularis (peroneus) longus and brevis
Subcutaneous calcaneal bursa
(Subtendinous) bursa of calcaneal tendon
Superior and Inferior fibular (peroneal) retinacula
Calcaneus
Extensor digitorum brevis muscle
Abductor digiti minimi muscle
Extensor digitorum longus muscle
Superior extensor retinaculum
Tendinous sheath of tibialis anterior
Lateral malleolus of fibula and subcutaneous bursa
Inferior extensor retinaculum
Tendinous sheath of extensor digitorum longus and fibularis (peroneal) tertius
Tendinous sheath of extensor hallucis longus
Fibularis (peroneus) longus tendon
Fibularis (peroneus) brevis tendon
Fibularis (peroneus) tertius tendon
Tuberosity of 5th metatarsal bone



Medial view
Tibialis anterior tendon and sheath
Tibia
Sheath of tibialis posterior tendon
Superior extensor retinaculum
Medial malleolus and subcutaneous bursa
Inferior extensor retinaculum
Tibialis posterior tendon and sheath
Tibialis anterior tendon and sheath
Tendinous sheath of extensor hallucis longus
1st metatarsal bone
Tendinous sheath of flexor hallucis longus
Medial plantar nerve and artery
Tendinous sheath of flexor digitorum longus
Flexor digitorum brevis muscle (cut)
Plantar aponeurosis (cut)
Abductor hallucis muscle (cut)
Lateral plantar nerve and artery
Calcaneus
Flexor retinaculum
(Subtendinous) bursa of calcaneal tendon
Subcutaneous calcaneal bursa
Tendinous sheath of flexor hallucis longus
Posterior tibial artery and tibial nerve
Tendinous sheath of flexor digitorum longus
Calcaneal (Achilles) tendon


Lateral view
Fibula
Tibia
Trochlea of talus
Posterior process of talus
Head of talus
Navicular
Sustentaculum tali of calcaneus
Lateral cuneiform
Calcaneus
Cuboid
Tuberosity of calcaneus
Tuberosity of
5th metatarsal
Medial calcaneal tubercle


Anterior view
Tibia
Fibula
Medial malleolus
Talus
Lateral malleolus

Physical Exam

- Inspection
- Palpation
  - Medial Malleolus
  - Posterior Heel
  - Peroneal Tendons
  - Anterior Ankle
- Range of Motion
  - Active- Already Assessed with squat
  - Passive
- Muscle strength

- Provocative Tests
  - Pott's Compression Test
  - Heel Tap Test
  - Anterior Drawer Test
  - Varus Stress Test (Talar Tilt)
  - Thompson Test

Inspection

Anterior View

OBSERVE THE ALIGNMENT OF THE GREATER AND LESSER TOES, THE POSITION OF THE FOOT IN RELATION TO THE LIMB, AND THE MEDIAL CURVATURE OF THE FOREFOOT.

Posterior View

ASSESS HEEL ALIGNMENT. NORMAL ALIGNMENT IS NEUTRAL OR SLIGHT VALGUS WITH NO MORE THAN 2 LATERAL TOES VISIBLE FROM BEHIND. ACQUIRED FLATFOOT WILL HAVE INCREASED VALGUS OF THE CALCANEUS AND MORE THAN 2 TOES WILL BE VISIBLE (“TOO MANY TOES” SIGN).

Inspection

Medial View

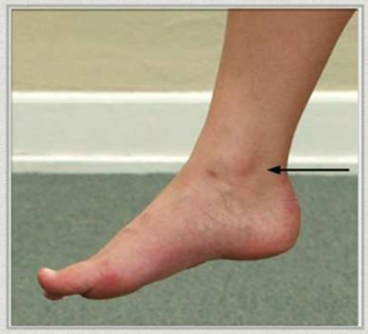
INSPECT FOR HIGH ARCH (CAVUS FOOT), FLATFOOT POSTURE (PES PLANS, OR UNDUE PROMINENCE OF THE MEDIAL MID FOOT (ACCESSORY NAVICULAR).

Lateral View

INSPECT FOR CALLOSITIES, ANKLE SWELLING, OR PROMINENCE OF THE POSTERIOR CALCANEUS.

Palpation: Medial Malleolus

- Palpate the area posterior and inferior to the medial malleolus in the region of the tibial nerve.
- Percussion over the nerve will reproduce "shooting" pain in patients with tarsal tunnel syndrome.
- Patients with posterior tibial tendon dysfunction will have swelling and tenderness along the course of the tendon.



Palpation: Posterior Heel

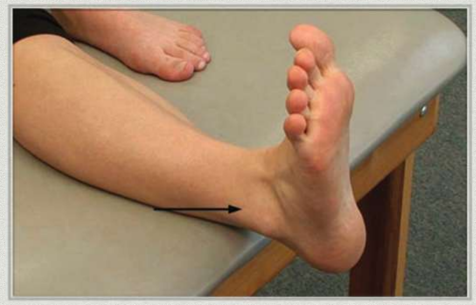
- Palpate both sides of the Achilles tendon insertion looking for swelling or tenderness which are signs of retrocalcaneal bursitis.
- Swelling and tenderness of the Achilles tendon at its insertion is associated with tendinitis or calcific tendinosis.

Palpation: Peroneal Tendons

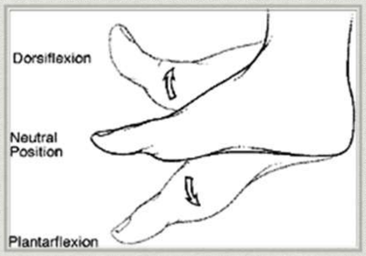
- Palpate behind and below the lateral malleolus for tenderness or swelling which is associated with peroneal tenosynovitis
- Palpate for subluxation of the tendons during active eversion, dorsiflexion, and plantar flexion.

Palpation: Anterior Ankle

- Palpate over the anterior talofibular ligament then then calcaneofibular ligament for tenderness associated with a sprain from acute inversion.



Passive Range of Motion

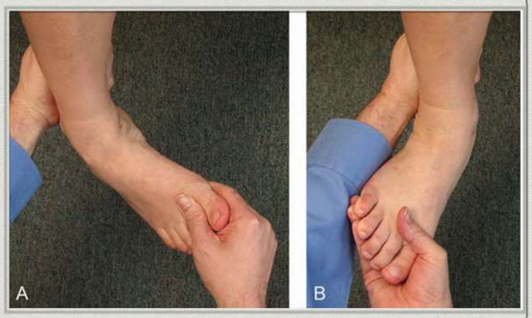


CHECK DORSIFLEXION AND PLANTAR FLEXION STARTING FROM A NEUTRAL POSITION

CHECK INVERSION (HEEL TURNING INWARD) AND EVERSION (HEEL TURNING OUTWARD). THESE ARE THE PRIMARY MOTIONS AT THE SUBTALAR JOINT.

Passive Range of Motion

- Supination and pronation refer to rotation of the foot about an AP axis
- Supination (A) includes inversion of the heel, as well as adduction and plantar flexion of the mid foot
- Pronation (B) includes eversion of the heel and abduction and dorsiflexion of the mid foot.



Muscle Testing: Posterior Tibialis

- Stabilize the lateral aspect of the leg with the foot in plantar flexion. This eliminates the activity of tibialis anterior
- Place the other hand on the medial aspect of the foot and ask the pt to invert the foot
- Weakness may indicate injury or dysfunction of the posterior tibialis, posterior tibial nerve or L5 nerve root

Muscle Testing: Anterior Tibialis

- Grasp the posterior lateral aspect of the leg
- Apply resistance to the dorsal medial aspect of the foot
- Ask the patient to flex the toes (to eliminate activity of the toe extensors) then invert and dorsiflex the foot against resistance
- Weakness indicates a lesion involving the L4 nerve root or deep peroneal nerve

Pott's Compression Test

- Evaluates for the presence of a fracture of the lower leg
- Examiner places the pads of the hands on either side of the upper portion of the leg then squeezes the fibula and tibia together
- Abnormal test: pain in the distal tibia or fibula may indicate a fracture

Squeeze Test

- Compress the tibia and fibula above the ankle to assess for a syndesmotic sprain (high ankle sprain)

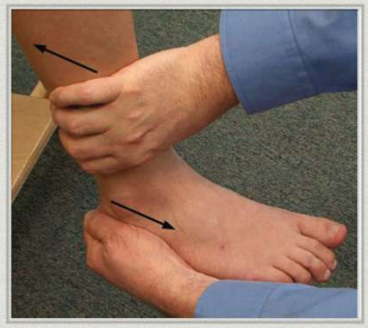
Squeeze Test Video

Heel Tap Test

- Patient is supine with foot of the involved leg off the table and knee straight
- Examiner stabilizes the lower leg with one hand and bumps the calcaneus with the other hand
- Abnormal test: pain is felt in the area of complaint. Indicates a possible fracture.

Anterior Drawer Test

- Assesses the stability of the **anterior talofibular ligament**
- With the patient seated and the knee flexed to 90º, place the ankle in approximately 20º of plantar flexion
- Stabilize the anterior aspect of the distal tibia
- Cup the palm of the other hand around the posterior aspect of the calcaneus
- Translate the calcaneus and talus forward on the tibia
- Abnormal test will have an absence of a firm end point and asymmetric or excessive motion



[Anterior Drawer Test Video]

Varus Stress Test

- Also called the Talar Tilt Test
- With the tibia stabilized and the ankle in neutral, grasp the calcaneus and invert the hind foot
- Excessive or asymmetric motion will occur with chronic laxity of the calcaneofibular ligament

[Varus Stress Test Video]

Thompson Test

- With the patient prone and relaxed, use your fingers and thumb to squeeze the medial and lateral aspect of the mid calf together
- Normal test: the foot moves into plantar flexion
- Abnormal test: the foot does not move into plantar flexion, then the test is positive for an **Achilles tendon rupture**

Thompson Test Video

Ankle Exam Video

- [Bates Ankle Exam Video]